INVERTIBLE MODULES HAVE REGULAR MEROMORPHIC SECTIONS IN THREE CASES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. Cases (1) and (2) are argued in place; case (3) is a single named input, and that shape is recorded here for the reader.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Meromorphic functions and sections on an integral scheme. Lemma lemma-meromorphic-functions-integral-scheme, used at proof lines 6524-6528. Identifies the two sheaves with constant sheaves at the generic point.
- Description of K_X and K_X(L) for a reduced scheme with finitely many components. Lemma lemma-reduced-finite-irreducible (6424-6456, produced as S15-div-reduced-finite-irreducible), used at proof lines 6534-6537. Makes the product of the chosen nonzero elements a meromorphic section.
- Existence of a regular meromorphic section in the Noetherian case. Lemma lemma-regular-meromorphic-section-exists-noetherian, used at proof line 6541. Is case (3), cited by name.

Source reference. Original lemma 01X5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2506–2506. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02OW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3531–3531. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EMI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18308–18308. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. In each of the following cases $\mathcal{L}$ has a regular meromorphic section:*

(1) *X is integral,*
(2) *X is reduced and any quasi-compact open has a finite number of irreducible components,*
(3) *X is locally Noetherian and has no embedded points.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. In case (1) let $\eta \in X$ be the generic point. We have seen in Lemma 01X5 that $\mathcal{K}_X$, resp. $\mathcal{K}_X(\mathcal{L})$ is the constant sheaf with value $\kappa(\eta)$, resp. $\mathcal{L}_\eta$. Since $\dim_{\kappa(\eta)} \mathcal{L}_\eta = 1$ we can pick a nonzero element $s \in \mathcal{L}_\eta$. Clearly s is a regular meromorphic section of $\mathcal{L}$. In case (2) pick $s_\eta \in \mathcal{L}_\eta$ nonzero for all generic points η of X; this is possible as $\mathcal{L}_\eta$ is a 1-dimensional vector space over $\kappa(\eta)$. It follows immediately from the description of $\mathcal{K}_X$ and $\mathcal{K}_X(\mathcal{L})$ in Lemma 02OW that $s = \prod s_\eta$ is a regular meromorphic section of $\mathcal{L}$. Case (3) is Lemma 0EMI. □